

ME602T - Assignment 3

The program is required to be run in the format `.\assignment_3.exe a.dat b.dat .`

The standard main function arguments of `int argc` and `char** argv` are used to get the file names of `a.dat` and `b.dat`.

The data from `.dat` files are read using the `fstream` and `sstream` header files.

Matrix from each file is read using the `ReadFileToMat()` helper function, which takes a `char*` file name and `vector<vector<float>>` matrix reference as arguments and does the required checks to make sure the correct file has been opened properly.

Since the standard extraction operator `>>` skips all whitespaces including `\n` characters, each row in the matrix is read separately using `getline()` function and is sent into a string `line`. This string is then opened as a string stream and all values in the row are collected and pushed into a temporary `vector<float>` `temp`. Once the string is completely read, the `temp` vector is pushed into the matrix reference taken as function argument. This is repeated till the entire file is read. Once the file is completely read, the stream is then closed and the function returns control to the `main` function.

The matrices read from the `.dat` file are stored as row-major matrices, which are then converted to column-major matrices to facilitate column wise multiplication as done in Assignment-2. The helper function `ScaleAndAddVector()` is used to compute the linear combination of the columns.

The required checks are done to make sure the matrices can be multiplied, and the resulting matrix is stored in another matrix of the correct dimensions obtained from the input matrices. This matrix is written to `.dat` file using `WriteMatToFile()` helper function, which takes only a reference to the matrix. The output file is opened as a file stream again and each element is written to the file row-wise with a white space after every element and a new line character at the end of every row. At the end `Matrix saved successfully.` is printed to console and the program exits.

Example test cases and their outputs:

Case 1: Not enough arguments provided when the program is run

```
Input: .\assignment_3.exe a.dat
Output: Usage .\assignment_3.exe <a.dat> <b.dat>
```

Case 2: Wrong file name used in input

```
Input: .\assignment_3.exe d.dat b.dat
Output: Error reading file d.dat
```

Case 3: Incompatible matrix order

```
a.dat:  
1 2 3  
b.dat:  
5  
Input: .\assignment_3.exe a.dat b.dat  
Output: Matrices cannot be multiplied.
```

Case 4: Compatible matrices with correct usage

```
a.dat:  
1 2 3  
4 5 6  
b.dat:  
7 8  
9 10  
11 12  
Input: .\assignment_3.exe a.dat b.dat  
Output: Matrix saved successfully.  
c.dat:  
58 64  
139 154
```